LedgerForge — Phase 2 Report

WorkforceOS: Attendance, Payroll Engine, Payslip PDF, Double-Entry Ledger & Receivables

Candidate Name: Ancillin P G | **Track:** Python Developer
Phase: Phase 2: Core Development (Days 6 to 10 / Tasks LF-06 to LF-10) | **Git Tag:** phase-2
GitHub Repository: https://github.com/Ancilin/Foundation-Project-Setup-Employee-Data-Model

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1. Phase 2 Task Execution & Evidence Table (Days 6 to 10)

Day / Task	Objective & Scope	Status	Deliverables & Evidence Links
Day 6 LF-06	**Record Time: Attendance & Leave** Chunked 150,155-row attendance loader, monthly_summary(), MISSING_CHECKOUT rule, holiday/weekend exclusion, leave apply/approve/reject.	**DONE**	• Services: attendance_service.py, leave_service.py • Hand-Count Parity: 3 employees (EMP-0002, EMP-0011, EMP-0148) • Tests: tests/test_attendance_and_leave.py (4/4 passed)
Day 7 LF-07	**The Payroll Calculation Engine** calculate_employee_month() pure function: calendar-day LOP, 2x OT (50h cap), PF (12%, INR 15k ceiling), ESI, PT, New Regime TDS, Decimal ROUND_HALF_UP.	**DONE**	• Engine: app/services/payroll_service.py • Verifier: **7,340 / 7,340 = 100.00% exact paisa match** • Tests: tests/test_payroll_engine.py (9/9 passed)
Day 8 LF-08	**Produce the Payslip People Actually Read** GET /payslips/{employee_id}/{run_id} JSON & ReportLab PDF, Indian currency formatting (1,23,456.00), Net Pay in words, 403 stranger refusal.	**DONE**	• Service & Routes: payslip_service.py, routes/payslip.py • Sample PDFs: exports/payslips/payslip_EMP-0002_PR-2025-04.pdf • Tests: tests/test_payslips.py (5/5 passed, incl. 403 check)
Day 9 LF-09	**Keep the Books: Double-Entry Ledger** 47-account Chart of Accounts, create_journal_entry() refusing unbalanced entries, read-only POSTED entries, reversals, account_balance(), PR-2025-04 posting.	**DONE**	• Service: app/services/ledger_service.py • Check 1: debits-credits=0.0, unbalanced entries=0 • Tests: tests/test_ledger.py (4/4 passed, JE-000105 exact)
Day 10 LF-10	**Earn Money: Clients, Invoices & Payments** 28 clients, 743 invoices (18% domestic / 0% export GST), status lifecycle, DRAFT/CANCELLED non-posting guard, payment receipts, AR reconciliation.	**DONE**	• Service: app/services/invoice_service.py • Check 3: ledger 1100 = open invoices = 139,394,278.00 • Tests: tests/test_invoices.py (3/3 passed)

2. Mini Challenge: How Attendance Becomes a Payslip and Then a Ledger Entry (5 Lines + Real Proof)

Line 1 (Attendance to Payable Days): Daily check-ins and approved leaves in monthly_summary() determine an employee's calendar lop_days (absent + 0.5 x half-day unpaid) and eligible overtime_hours.
Line 2 (Fixed & Variable Earnings): calculate_employee_month() prorates each fixed component (BASIC, HRA, CONV, MED, SPECIAL) by (days_employed - lop_days) / calendar_days and adds overtime pay (2x hourly rate) and bonus to form gross_pay.
Line 3 (Statutory Deductions to Net Pay): Employee PF (12% of earned basic up to INR 15,000), ESI (0.75% if fixed gross <= INR 21,000), Professional Tax (INR 200 if gross >= INR 25,000), and progressive New-Regime TDS are subtracted from gross_pay to yield net_pay.
Line 4 (Engine to Payslip Document): get_payslip_data() and generate_payslip_pdf() render those exact engine figures into an owner-protected PDF formatted in the Indian numbering system (1,23,456.00).
Line 5 (Payslip to Double-Entry Ledger): post_payroll_run_to_ledger() debits departmental employer-cost accounts (5001-5005: Fixed Pay, OT, Bonus, Employer PF/ESI) and credits liability/recovery accounts (2100 Net Pay, 2110 PF, 2120 ESI, 2130 TDS, 2140 PT, 1300 Advances) so Total Debits equal Total Credits.

Real Employee Proof — EMP-0011 (Liam Ahuja, Dept D07, Role R039) in PR-2025-04 (April 2025):
1. **Attendance (2025-04):** 30 calendar days, 0.50 LOP days (1 half-day unpaid) -> **29.50 payable days**, plus **1.50 overtime hours**.
2. **Payslip Earnings:** Monthly Fixed Gross INR 24,041.00 prorated by 29.5/30 = INR 23,640.32 (Basic 9,455.73 + HRA 4,727.87 + Conv 1,573.33 + Med 1,229.17 + Special 6,654.22) + Overtime INR 346.75 + Bonus INR 12,982.14 = **Gross Pay INR 36,969.21**.
3. **Payslip Deductions & Net Pay:** Employee PF (12% of 9,455.73) = INR 1,134.69 + PT = INR 200.00 + TDS = INR 0.00 -> Total Deductions = **INR 1,334.69** -> **Net Pay = INR 35,634.52** (Employer PF = INR 1,134.69; Total Employer Cost = **INR 38,103.90**).
4. **Ledger Entry (JE-000105):** Contributes INR 23,640.32 to Dr 5001 (D07), INR 346.75 to Dr 5002 (D07), INR 12,982.14 to Dr 5003 (D07), and INR 1,134.69 to Dr 5004 (D07) (Total Debit = **INR 38,103.90**), balanced by Cr 2100 (INR 35,634.52), Cr 2110 (INR 2,269.38), and Cr 2140 (INR 200.00) (Total Credit = **INR 38,103.90**).

3. Proof It Works: Automated Tests & Official Verifier Outputs

A. Official Payroll Engine Verifier & Ledger Integrity Checker (verify_ledgerforge.py)

```
$ python scripts/verify_ledgerforge.py payroll exports/my_payroll.csv --data data/csv
Rows expected 7340, rows found in your file 7340
Exact matches (within 1 paisa): 7340/7340 = 100.00%

$ python scripts/verify_ledgerforge.py checks --data exports/
PASS Every journal entry balances - debits-credits=0.0, unbalanced entries=0
PASS Payroll equals ledger, every month - largest monthly difference=0.00
PASS Receivables equal open invoices - ledger 1100=139,394,278.00, open (sent/overdue/part-paid) invoices=139,394,278.00
PASS Payroll employees exist - payroll rows with unknown employee=0
PASS Nobody is paid after leaving - payroll rows after exit_date=0
5/5 checks passed
```

B. Automated Pytest Suite (96 Passed, 0 Failed)

```
============================= test session starts =============================
platform win32 -- Python 3.12.10, pytest-8.4.2
collected 96 items

tests/test_attendance_and_leave.py (4 tests: 3-emp hand count, MISSING_CHECKOUT, holidays, leave) . PASSED
tests/test_payroll_engine.py (9 tests: LOP, OT, PF, ESI, PT, TDS, 5 real rows, Decimal vs float) .. PASSED
tests/test_payslips.py (5 tests: Indian format_inr, 403 stranger refusal, 3 PDFs vs CSV) .......... PASSED
tests/test_ledger.py (4 tests: 47 accounts, 422 unbalanced refusal, 409 immutable, JE-000105) ..... PASSED
tests/test_invoices.py (3 tests: Check 3 AR reconciliation, DRAFT/CANCELLED guard, payments) ...... PASSED
tests/test_migrations.py + Phase 1 suite (71 tests: Alembic zero-drift, CRUD, hierarchy, salary) .. PASSED

======================== 96 passed, 4 warnings in 179.97s ========================
```

4. What I Learned in Phase 2 (In My Own Words)

Phase 2 taught me how operational HR events, payroll calculations, and double-entry accounting form a single unbroken chain where every rupee must reconcile. I learned why financial systems never use binary floating-point numbers and how Decimal with ROUND_HALF_UP prevents 1-paisa rounding drift across 7,340 payroll rows. I also learned how accountants protect audit history by making posted journal entries strictly read-only—correcting mistakes or cancelled invoices exclusively through reversing entries so the general ledger and business sub-ledgers never disagree.

5. Problems Encountered & How I Solved Them

1. 1-Paisa Rounding Boundary in Monthly TDS on Variable Pay (Task 7)

Problem: When combining regular monthly prorated TDS and one-time bonus TDS before rounding, 106 out of 7,340 rows differed from payroll_items.csv by INR 0.01.
Solution: Diagnosed the exact order of operations: prorating regular monthly tax by payable_days / calendar_days and rounding regular TDS and marginal bonus TDS separately with ROUND_HALF_UP achieved **7,340 / 7,340 (100.00%)** exact matches.

2. Handling Imperfect Biometric Data (MISSING_CHECKOUT) Without Corrupting Overtime (Task 6)

Problem: Rows flagged MISSING_CHECKOUT had a valid check_in but null check_out and null hours_worked, which risked crashing numeric sums or inflating hours.
Solution: Established a deterministic written rule: count the day as PRESENT (no unfair LOP), default hours_worked and overtime_hours to 0.00, and surface missing_checkout_count in the monthly summary.

3. Preserving Check 3 (Receivables Reconciliation) When a Sent Invoice Is Cancelled (Task 10)

Problem: Cancelling a SENT invoice removes it from open invoices, which would cause Ledger Account 1100 to exceed open receivables if the original ledger entry remained unadjusted.
Solution: Built cancel_invoice() to automatically call reverse_journal_entry() whenever a posted invoice with zero payments is cancelled, debiting Revenue/GST and crediting Account 1100.

6. Sample Evidence Tables (Tasks 8, 9 & 10)

A. Sample Payroll Journal Entry Table (JE-000105 — PR-2025-04)

Account	Account Name	Cost Centers	Debit (INR)	Credit (INR)
5001	Salaries & Wages (Earned Fixed Pay)	D00–D10 (11 depts)	39,776,995.27	0.00
5002	Overtime Expense (2x Hourly)	D06, D07, D08, D09	170,928.18	0.00
5003	Incentives & Bonus (Variable Pay)	D01–D10 (10 depts)	12,792,186.98	0.00
5004	Employer PF Contribution (12%)	D00–D10 (11 depts)	618,963.94	0.00
5005	Employer ESI Contribution (3.25%)	D06	4,317.71	0.00
2100	Salaries Payable (Take-Home Net Pay)	—	0.00	46,035,111.16
2110 / 2120	PF Payable (1,237,927.88) & ESI Payable (5,314.10)	—	0.00	1,243,241.98
2130 / 2140 / 1300	TDS (6,007,238.94), PT (74,800.00), Advances (3,000.00)	—	0.00	6,085,038.94
TOTAL	**43 Balanced Lines (status = POSTED)**	**PR-2025-04**	**53,363,392.08**	**53,363,392.08**

7. Plan for Next Phase (Phase 3 — Days 11 to 15: Integration, Reports & Security)

Day 11 (LF-11) — Vendor Expenses & Bill Payments: Model vendor bills with GST Input Credit (Account 1200), expense categories (5101–5112), and vendor payments clearing Accounts Payable (2001).

Day 12 (LF-12) — End-to-End Payroll Run & Idempotent Posting: Automate full-company monthly payroll runs with idempotency keys (payroll:YYYY-MM:ZNT), linking each run directly to its posted journal entry and bank disbursement.

Day 13 (LF-13) — Authentication & Role-Based Access Control (RBAC): Implement JWT authentication and role guards (ADMIN, HR, FINANCE, MANAGER, EMPLOYEE) across all endpoints with an audit log.

Day 14 & 15 (LF-14 & LF-15) — Trial Balance, P&L, Balance Sheet & Final Verification: Build the Trial Balance (verified via verify_ledgerforge.py trial-balance), Profit & Loss statement, Balance Sheet, and cash-flow summary.

8. Submission & Git Tag Sign-Off

Candidate Signature: Ancillin P G | **Date:** October 26, 2026
Git Remote URL: https://github.com/Ancilin/Foundation-Project-Setup-Employee-Data-Model
Git Tags Verified: phase-1, phase-2, v0.0.1